“Two Voices, Two Perspectives”: Men and Women in Movie Dialogues

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Abstract

There has been extensive research into whether men and women speak differently across different contexts [1, 2, 3]. My project sought to answer this question in the context of movie dialogue. I extracted data from Cornell Movie-Dialogs Corpus and analyzed how dialogue sentiments differ across different gender groupings over multiple decades. I found that 1) women have significantly higher dialogue sentiments; 2) sentiments of dialogue between same-gender speakers are significantly lower than between opposite-gender speakers. I conclude that these speech sentiment differences have been consistent over time, which may stem from social implications related to gender and how we speak to in-group versus out-group people.

Research Questions

1. Has movie dialogue sentiment changed over time?
2. Has the sentiment of movie dialogue spoken by men versus women changed over time?
3. Has the sentiment of movie dialogue spoken by men to men, by men to women, by women to men, and by women to women changed over time?

Dataset Description

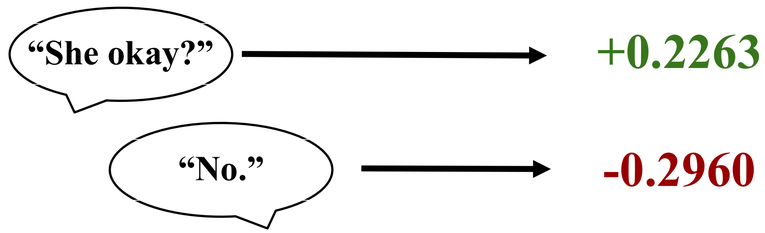
I used the **Cornell Movie--Dialogs Corpus**, a collection of fictional conversations extracted from raw movie scripts. It includes 220,579 conversational exchanges between 10,292 pairs of movie characters, with 9,035 characters from 617 movies. In total, I analyzed 304,713 “utterances” (a single line spoken by a character). I extracted the following info using the ConvoKit package on the corpus’ utterance, speaker, and conversation classes, and joined data gathered from each:

- Utterance text
- Utterance ID
- Response utterance ID
- Speaker ID
- Speaker gender
- Movie name
- Movie release year

I excluded utterances from speakers with unknown genders and categorized movies according to release decade.

Sentiment Analysis

I used NLTK’s Pre-Trained Sentiment Analyzer, which is well-suited for short sentences. For each utterance, it calculates a compound sentiment score between +1 (extremely positive sentiment) and -1 (extremely negative sentiment).



Data Exploration

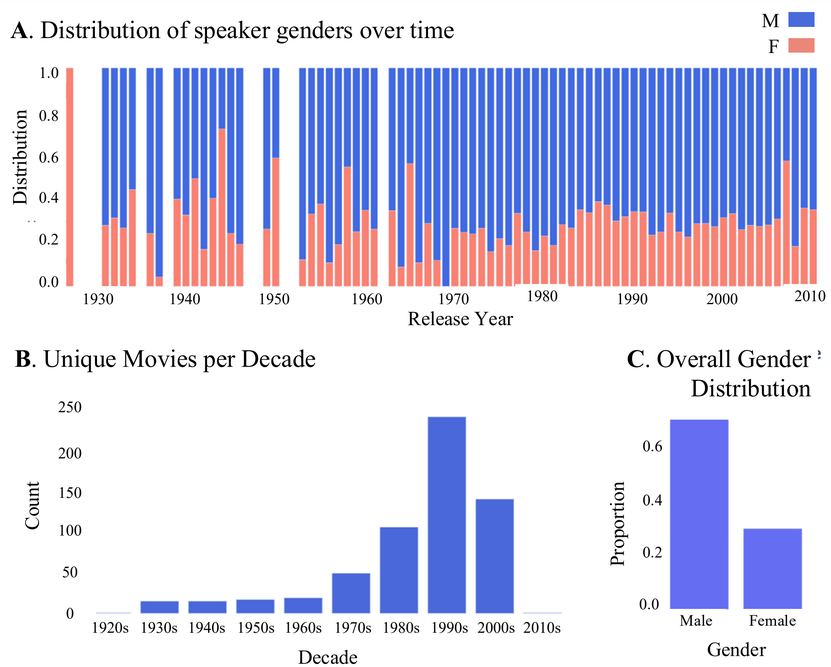


Figure 1. Exploratory analysis of movie corpus. A) After filtering utterances with unknown speaker gender, a larger proportion of utterances are consistently made by men. Years with no information are years where no utterance speaker genders were known. **B)** Before the 1970s, there are not a significant number of unique movies in the corpus. **C)** Overall, a larger proportion of utterances in the corpus are spoken by men.

Data Analysis

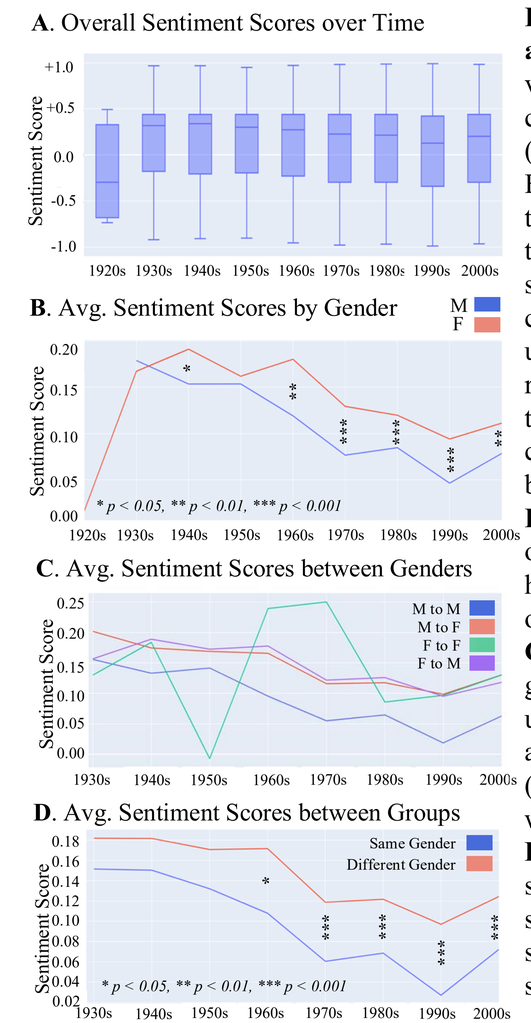


Figure 2. Sentiment score analysis. A TF-IDF vectorizer was fit to the corpus of all utterances (1000 maximum features). Each utterance was filtered to include only words in these features. Sentiment score (SS) was then calculated for each utterance. **A)** SS patterns remain consistent over time, with no strong correlation (r = -0.065) between score and year. **B)** T-test of SS by gender over time show that women have statistically higher SS over multiple decades. **C)** Average SS between genders show male-to-male utterances with lower SS and within-gender group (M to M, F to F) utterances with higher SS. **D)** T-test of SS between same speaker genders are significantly higher than scores between opposite speaker genders.

Conclusion

My results show that while overall movie dialogue sentiment has not changed significantly over time, trends in sentiment differences across genders and in conversations between different gender groups have been consistent. However, My results are limited by the dataset’s imbalanced speaker gender ratio and the imbalanced number of movies per decade. Future analysis on this dataset might explore how dialogue sentiment overall and in conversations between genders varies with movie rating or genre.

[1] Tripp, A., & Munson, B. (2022). Perceiving gender while perceiving language: Integrating psycholinguistics and gender theory. Wiley Interdisciplinary Reviews: Cognitive Science, 13(2), e1583.
[2] Charlesworth, T. E. et al.(2021). Gender stereotypes in natural language: Word embeddings show robust consistency across child and adult language corpora of more than 65 million words. Psychological Science, 32(2), 218-240.
[3] Stokoe, E. (2018, November 8). Do women and men talk differently?. BPS. https://www.bps.org.uk/psychologist/do-women-and-men-talk-differently.